

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12399105
B2
August 26, 2025
Wu; Dacheng et al.

Structural intelligent force-measurement apparatus and in-situ calibration method

Abstract

A method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction includes steps of A1: recording a value of a horizontal or lateral force applied to a force measurement apparatus, and recording a horizontal or lateral force value of a sensing apparatus (5) before the force measurement apparatus is lifted; then applying a horizontal or lateral force to two wedge-shaped load transducers (3), and recording a horizontal or lateral force value of the sensing apparatus (5) during the change in height or during a pause in the change; A2: deriving a relational expression of a coefficient of friction, an angle of inclination of the load transducers (3) and a horizontal or lateral force; and A3: calculating a value of the coefficient of friction according to the recorded horizontal or lateral force values and the relational expression in A2.

Inventors:	Wu; Dacheng (Sichuan, CN), Wang; Jianming (Sichuan, CN), Zou; Yijun (Sichuan, CN), Deng; Yu (Sichuan, CN)
Applicant:	SICHUAN SUNLIGHT INSPECTION AND TESTING CO., LTD. (Sichuan, CN)
Family ID:	1000008781205
Assignee:	Wu; Dacheng (Chengdu, CN)
Appl. No.:	18/033348
Filed (or PCT Filed):	March 18, 2021
PCT No.:	PCT/CN2021/081402
PCT Pub. No.:	WO2022/083054
PCT Pub. Date:	April 28, 2022

Prior Publication Data

Document Identifier

US 20230393053 A1

Publication Date

Dec. 07, 2023

Foreign Application Priority Data

CN	202011148106.8	Oct. 23, 2020
CN	202023075458.3	Dec. 18, 2020
CN	202011538900.3	Dec. 23, 2020
CN	202011538911.1	Dec. 23, 2020

Publication Classification

Int. Cl.: **G01N19/00** (20060101); **E01D19/04** (20060101); **G01L25/00** (20060101); **G01N19/02** (20060101)

U.S. Cl.:

CPC **G01N19/02** (20130101); **G01L25/00** (20130101);

Field of Classification Search

CPC: G01N (3/08); G01N (3/32); G01N (3/20); G01N (3/04); G01N (3/00); G01N (19/04); G01N (2203/0282); G01N (3/10); G01N (2203/0023); G01N (2203/0017); G01N (3/24); G01N (2203/0073); G01N (2203/028); G01N (3/02); G01N (2203/0016); G01N (2203/0278); G01N (2203/0222); G01N (2203/0019); G01N (2291/02827); G01N (2203/0226); G01N (3/06); G01N (2203/0296); G01N (33/383); G01N (3/14); G01N (2203/0005); G01N (3/12); G01N (2203/0025); G01N (33/24); G01N (3/34); G01N (2203/0066); G01N (3/42); G01N (3/18); G01N (2203/027); G01N (29/14); G01N (2203/0405); G01N (2203/0623); G01N (2203/0048); G01N (3/56); G01N (3/066); G01N (3/40); G01N (2203/0071); G01N (2203/0274); G01N (3/38); G01N (3/36); G01N (3/22); G01N (2203/0021); G01N (2203/0033); G01N (2203/0246); G01N (3/068); G01N (2203/0075); G01N (2203/0682); G01N (2203/0037); G01N (3/30); G01N (33/388); G01N (2203/0067); G01N (2203/0676); G01N (2203/0286); G01N (2203/0256); G01N (2203/0244); G01N (2203/0094); G01N (3/16); G01N (2203/0254); G01N (2203/0647); G01N (2203/0284); G01N (2203/0062); G01N (2203/0617); G01N (2203/0051); G01N (33/42); G01N (2203/021); G01N (2203/0089); G01N (2203/0298); G01N (29/227); G01N (2203/0085); G01N (2203/0208); G01N (29/07); G01N (2203/0091); G01N (2203/0218); G01N (2203/0092); G01N (2203/0064); G01N (2203/0042); G01N (2203/0435); G01N (2203/0039); G01N (19/08); G01N (2203/005); G01N (1/286); G01N (3/303); G01N (17/00); G01N (2203/0044); G01N (2291/0231); G01N (33/346); G01N (33/365); G01N (2203/0411); G01N (3/28); G01N (19/02); G01N (2203/0026); G01N (11/00); G01N (11/14); G01N (2203/04); G01N (33/442); G01N (2203/0292); G01N (3/062); G01N (2203/0003); G01N (33/46); G01N (33/367); G01N (2203/024); G01N (2203/0664); G01N (33/386); G01N (2203/0288); G01N (2291/044); G01N (29/11); G01N (33/445); G01N (3/60); G01N (3/565); G01N (2203/001); G01N (2291/2691); G01N (33/0078); G01N (2203/0447); G01N (2203/0035); G01N (2203/0087); G01N (2203/0098); G01N (2203/0266); G01N (2291/0258); G01N

(2203/0641); G01N (29/12); G01N (2203/0258); G01N (2203/0688); G01N (2203/0007); G01N (27/72); G01N (11/165); G01N (29/2418); G01N (3/62); G01N (33/38); G01N (2203/0242); G01N (33/34); G01N (33/44); G01N (2203/0028); G01N (2291/02854); G01N (2291/0422); G01N (2203/0252); G01N (27/205); G01N (2203/0055); G01N (2203/0206); G01N (2291/0421); G01N (33/36); G01N (2203/0224); G01N (3/26); G01N (19/00); G01N (2203/0248); G01N (2203/0476); G01N (33/008); G01N (2203/0423); G01N (29/045); G01N (29/2412); G01N (33/15); G01N (2203/0658); G01N (11/142); G01N (2203/0494); G01N (2203/0216); G01N (27/20); G01N (2203/0212); G01N (2291/011); G01N (2203/0008); G01N (2203/0214); G01N (2203/0268); G01N (2291/02872); G01N (27/725); G01N (29/223); G01N (3/307); G01N (11/10); G01N (2203/0032); G01N (2203/0264); G01N (2291/2694); G01N (29/30); G01N (3/48); G01N (3/58); G01N (2203/006); G01N (2203/0082); G01N (2291/0427); G01N (33/20); G01N (33/4833); G01N (2203/0694); G01N (11/162); G01N (13/00); G01N (2203/0441); G01N (2203/0482); G01N (25/72); G01N (29/043); G01N (2203/0041); G01N (2203/0635); G01N (2291/048); G01N (2291/106); G01N (2203/0057); G01N (2203/0452); G01N (2203/0605); G01N (33/0081); G01N (33/0091); G01N (9/36); G01N (11/16); G01N (2203/0053); G01N (2203/0069); G01N (2203/0228); G01N (17/02); G01N (2291/101); G01N (2291/102); G01N (27/24); G01N (33/02); G01N (15/0826); G01N (19/10); G01N (2203/0429); G01N (2291/015); G01N (2291/0423); G01N (25/16); G01N (29/2475); G01N (3/319); G01N (2291/0232); G01N (33/32); G01N (17/006); G01N (2203/0012); G01N (2203/0096); G01N (203/0276); G01N (29/4427); G01N (29/46); G01N (3/064); G01N (21/21); G01N (2203/0076); G01N (2291/267); G01N (29/2437); G01N (2203/008); G01N (2203/0234); G01N (2291/0251); G01N (2291/2695); G01N (3/165); G01N (33/10); G01N (33/26); G01N (11/04); G01N (15/08); G01N (21/88); G01N (2203/0204); G01N (2291/02881); G01N (27/223); G01N (29/348); G01N (33/222); G01N (33/54373); G01N (2291/0237); G01N (2291/0289); G01N (3/313); G01N (3/567); G01N (33/2045); G01N (35/00871); G01N (2203/0464); G01N (23/207); G01N (29/4418); G01N (3/46); G01N (33/0083); G01N (15/0806); G01N (2203/0629); G01N (2291/0238); G01N (2291/02863); G01N (29/24); G01N (29/343); G01N (33/0077); G01N (33/0086); G01N (2203/0236); G01N (2203/0272); G01N (27/9046); G01N (29/222); G01N (3/31); G01N (3/405); G01N (5/04); G01N (11/02); G01N (21/95); G01N (2203/023); G01N (2203/0232); G01N (2203/029); G01N (2203/0417); G01N (2291/014); G01N (2291/0235); G01N (2291/103); G01N (23/046); G01N (27/82); G01N (33/204); G01N (15/088); G01N (21/8806); G01N (21/91); G01N (2203/06); G01N (2291/2623); G01N (2291/2693); G01N (23/20); G01N (25/00); G01N (27/041); G01N (29/42); G01N (29/4472); G01N (33/0003); G01N (33/241); G01N (1/04); G01N (13/02); G01N (2013/0216); G01N (2203/0238); G01N (2203/026); G01N (2223/419); G01N (2291/02416); G01N (2291/02466); G01N (2291/105); G01N (2291/2634); G01N (29/04); G01N (29/069); G01N (29/28); G01N (33/025); G01N (33/12); G01N (21/1702); G01N (2203/0262); G01N (2203/047); G01N (2291/02836); G01N (27/04); G01N (29/036); G01N (29/326); G01N (29/4436); G01N (33/00); G01N (1/28); G01N (1/32); G01N (21/01); G01N (21/70); G01N (21/952); G01N (2203/0078); G01N (2203/0652); G01N (2291/02491); G01N (2291/265); G01N (25/04); G01N (27/90); G01N (29/022); G01N (29/4445); G01N (33/0096); G01N (33/30); G01N (33/447); G01N (35/00); G01N (5/02); G01N (17/002); G01N (2001/021); G01N (2021/8472); G01N (21/1717); G01N (21/8422); G01N (2203/0046); G01N (2203/0458); G01N (2291/2675); G01N (2291/269); G01N (27/22); G01N (27/902); G01N (29/024); G01N (29/041); G01N (29/048); G01N (29/22); G01N (29/228); G01N (29/245); G01N (29/262); G01N (29/4463); G01N (29/449); G01N (3/52); G01N (3/54); G01N (33/48); G01N (33/54346); G01N (35/04); G01N

(5/00); G01N (15/00); G01N (17/043); G01N (2015/0096); G01N (21/64); G01N (2203/0202); G01N (2291/02441); G01N (2291/0256); G01N (2291/02818); G01N (2291/2672); G01N (27/904); G01N (27/9053); G01N (29/09); G01N (29/221); G01N (29/36); G01N (3/44); G01N (33/0095); G01N (33/2823); G01N (7/10); G01N (9/00); G01N (1/00); G01N (11/08); G01N (15/02); G01N (15/082); G01N (17/046); G01N (19/06); G01N (2001/2873); G01N (2013/006); G01N (2015/0846); G01N (2035/00881); G01N (2035/0441); G01N (21/211); G01N (21/31); G01N (21/954); G01N (2203/003); G01N (2203/025); G01N (2203/0611); G01N (2291/0428); G01N (2291/2626); G01N (2291/2636); G01N (23/083); G01N (23/2251); G01N (25/18); G01N (27/9006); G01N (29/0672); G01N (29/225); G01N (29/2481); G01N (29/4454); G01N (29/48); G01N (29/50); G01N (33/2022); G01N (35/0099); G01N (35/028); G01N (9/02); G01N (17/04); G01N (2001/025); G01N (2001/366); G01N (2011/0033); G01N (2015/0873); G01N (2015/1006); G01N (2015/1495); G01N (2015/0497); G01N (2021/1725); G01N (21/171); G01N (21/43); G01N (21/7703); G01N (21/8803); G01N (21/9501); G01N (21/9508); G01N (21/958); G01N (2201/0233); G01N (2203/0083); G01N (2203/022); G01N (2291/0234); G01N (2291/0255); G01N (2291/2632); G01N (23/18); G01N (24/081); G01N (25/20); G01N (25/4846); G01N (27/048); G01N (27/121); G01N (27/225); G01N (29/00); G01N (29/02); G01N (29/032); G01N (29/0681); G01N (29/075); G01N (29/265); G01N (29/323); G01N (29/44); G01N (3/50); G01N (33/0009); G01N (33/04); G01N (33/343); G01N (33/487); G01N (33/49); G01N (33/4905); G01N (35/0098); G01N (35/02); G01N (1/2202); G01N (1/2226); G01N (1/24); G01N (1/2813); G01N (15/04); G01N (2001/2223); G01N (2001/2866); G01N (2011/0026); G01N (2015/0038); G01N (2015/0092); G01N (2035/00772); G01N (21/0303); G01N (21/0317); G01N (21/15); G01N (21/25); G01N (21/27); G01N (21/47); G01N (21/8915); G01N (2201/06113); G01N (2203/00); G01N (2203/067); G01N (2291/0226); G01N (2291/02845); G01N (2291/0426); G01N (2291/056); G01N (2291/2698); G01N (23/04); G01N (23/203); G01N (24/08); G01N (24/085); G01N (24/087); G01N (25/02); G01N (25/12); G01N (25/4866); G01N (27/02); G01N (27/023); G01N (27/026); G01N (27/12); G01N (27/62); G01N (29/06); G01N (29/0609); G01N (29/2462); G01N (29/38); G01N (29/4409); G01N (31/22); G01N (33/009); G01N (33/1873); G01N (33/362); G01N (33/40); G01N (33/48735); G01N (35/025); G01N (5/025); G01N (9/24); G01N (1/02); G01N (1/08); G01N (1/18); G01N (1/2035); G01N (1/22); G01N (11/12); G01N (11/167); G01N (15/06); G01N (15/10); G01N (15/1031); G01N (15/1425); G01N (15/1433); G01N (15/1456); G01N (15/1468); G01N (15/147); G01N (2001/022); G01N (2001/028); G01N (2001/388); G01N (2011/002); G01N (2011/0066); G01N (2011/008); G01N (2011/0086); G01N (2013/0225); G01N (2013/0233); G01N (2021/036); G01N (2021/0389); G01N (2021/1706); G01N (2021/1708); G01N (2021/1787); G01N (2021/3595); G01N (2021/5969); G01N (2021/646); G01N (2021/7786); G01N (2021/7796); G01N (2021/8427); G01N (2021/8438); G01N (2021/8444); G01N (2021/8822); G01N (2021/8829); G01N (2021/8838); G01N (2021/8874); G01N (2021/8887); G01N (2035/00039); G01N (2035/00108); G01N (2035/00336); G01N (2035/00376); G01N (2035/00683); G01N (2035/00782); G01N (2035/00841); G01N (2035/00851); G01N (2035/0091); G01N (21/17); G01N (21/23); G01N (21/251); G01N (21/255); G01N (21/29); G01N (21/33); G01N (21/3581); G01N (21/45); G01N (21/453); G01N (21/57); G01N (21/636); G01N (21/6428); G01N (21/714); G01N (21/78); G01N (21/84); G01N (21/8483); G01N (21/8851); G01N (21/93); G01N (21/9503); G01N (21/9505); G01N (21/956); G01N (21/95684); G01N (22/02); G01N (22/04); G01N (2201/102); G01N (2201/13); G01N (2223/053); G01N (2223/09); G01N (2223/6116); G01N (2223/616); G01N (2291/018); G01N (2291/0224); G01N (2291/02408); G01N (2291/02433); G01N (2291/0245); G01N (2291/025); G01N (2291/104); G01N

(2291/2638); G01N (2291/2696); G01N (2291/2697); G01N (23/06); G01N (27/002); G01N (27/07); G01N (27/126); G01N (27/18); G01N (27/221); G01N (27/226); G01N (27/227); G01N (27/228); G01N (27/3271); G01N (27/4067); G01N (27/4071); G01N (27/4073); G01N (27/4145); G01N (27/42); G01N (27/447); G01N (27/44743); G01N (27/44773); G01N (27/74); G01N (27/83); G01N (27/84); G01N (27/85); G01N (27/9013); G01N (29/0645); G01N (29/0663); G01N (29/2406); G01N (29/2425); G01N (29/2456); G01N (29/26); G01N (29/27); G01N (29/34); G01N (29/346); G01N (29/40); G01N (29/4481); G01N (29/52); G01N (3/562); G01N (31/221); G01N (33/0027); G01N (33/0031); G01N (33/0085); G01N (33/0098); G01N (33/08); G01N (33/18); G01N (33/205); G01N (33/208); G01N (33/246); G01N (33/4875); G01N (33/54353); G01N (33/54366); G01N (33/569); G01N (33/6848); G01N (35/00009); G01N (35/00712); G01N (7/00); G01N (9/04); G01N (9/08); G01N (9/10); G01N (9/12); G01N (9/20); G01N (1/2208); G01N (1/2211); G01N (1/2806); G01N (1/36); G01N (1/38); G01N (1/4022); G01N (1/42); G01N (1/44); G01N (13/04); G01N (15/01); G01N (15/0266); G01N (15/0606); G01N (2001/383); G01N (2001/386); G01N (2001/4033); G01N (2013/0283); G01N (2015/019); G01N (2015/0261); G01N (2015/1022); G01N (2021/1704); G01N (2021/213); G01N (2021/4166); G01N (2021/434); G01N (2021/9518); G01N (2030/322); G01N (2035/00188); G01N (2035/00198); G01N (2035/00217); G01N (21/00); G01N (21/41); G01N (21/4133); G01N (2203/0058); G01N (2203/02); G01N (2223/04); G01N (2223/056); G01N (2223/1016); G01N (2223/108); G01N (2223/309); G01N (2223/311); G01N (2223/321); G01N (2223/418); G01N (2223/607); G01N (2223/62); G01N (2223/632); G01N (2223/646); G01N (2223/649); G01N (2291/0254); G01N (2291/043); G01N (23/02); G01N (23/201); G01N (25/68); G01N (27/00); G01N (27/127); G01N (29/0618); G01N (29/2431); G01N (29/2487); G01N (31/005); G01N (33/0006); G01N (33/0093); G01N (33/207); G01N (33/22); G01N (33/245); G01N (33/2841); G01N (33/39); G01N (33/574); G01N (33/00594); G01N (35/00603); G01N (5/045); G01L (1/18); G01L (1/125); G01L (1/2287); G01L (5/24); G01L (1/2206); G01L (1/16); G01L (1/225); G01L (1/2281); G01L (1/2293); G01L (1/24); G01L (1/20); G01L (1/22); G01L (1/142); G01L (1/127); G01L (5/0033); G01L (1/14); G01L (5/0047); G01L (1/26); G01L (5/0004); G01L (1/242); G01L (1/241); G01L (1/246); G01L (1/2231); G01L (1/2262); G01L (5/0076); G01L (5/0038); G01L (1/005); G01L (5/00); G01L (1/2218); G01L (5/246); G01L (3/102); G01L (5/243); G01L (1/205); G01L (9/0054); G01L (1/255); G01L (5/102); G01L (3/105); G01L (5/0052); G01L (5/10); G01L (5/162); G01L (5/0061); G01L (5/0042); G01L (9/0042); G01L (1/2268); G01L (9/0055); G01L (25/00); G01L (9/065); G01L (1/2243); G01L (5/0019); G01L (1/144); G01L (5/101); G01L (1/04); G01L (1/162); G01L (5/103); G01L (5/0057); G01L (5/1627); G01L (1/2225); G01L (1/247); G01L (1/00); G01L (5/165); G01L (3/10); G01L (1/2212); G01L (3/108); G01L (5/0085); G01L (5/107); G01L (9/0052); G01L (5/161); G01L (1/10); G01L (1/12); G01L (11/02); G01L (1/02); G01L (1/06); G01L (1/106); G01L (1/148); G01L (11/025); G01L (5/228); G01L (1/146); G01L (1/165); G01L (1/245); G01L (9/0002); G01L (19/147); G01L (5/06); G01L (1/2237); G01L (5/04); G01L (5/047); G01L (9/0073); G01L (9/0075); G01L (1/2256); G01L (5/225); G01L (9/0072); G01L (1/243); G01L (25/003); G01L (9/0051); G01L (1/248); G01L (19/0092); G01L (5/0028); G01L (5/042); G01L (5/045); G01L (5/108); G01L (5/223); G01L (1/25); G01L (5/008); G01L (5/167); G01L (9/045); G01L (9/16); G01L (3/12); G01L (5/106); G01L (5/12); G01L (9/0022); G01L (9/0098); G01L (9/06); G01L (1/183); G01L (19/04); G01L (1/2275); G01L (5/22); G01L (5/226); G01L (9/04); G01L (5/16); G01L (5/0071); G01L (5/136); G01L (19/148); G01L (9/0019); G01L (19/02); G01L (5/0009); G01L (9/002); G01L (3/103); G01L (3/104); G01L (5/28); G01L (9/007); G01L (9/12); G01L (1/186); G01L (19/0084); G01L (23/18);

G01L (5/105); G01L (19/086); G01L (3/242); G01L (5/221); G01L (9/0001); G01L (9/0061); G01L (1/044); G01L (1/122); G01L (5/0066); G01L (5/08); G01L (9/125); G01L (9/0035); G01L (1/042); G01L (23/10); G01L (3/101); G01L (19/0038); G01L (3/24); G01L (5/0014); G01L (5/009); G01L (5/133); G01L (5/164); G01L (9/006); G01L (9/0077); G01L (9/008); G01L (9/08); G01L (13/025); G01L (3/106); G01L (3/1457); G01L (5/14); G01L (19/0636); G01L (19/146); G01L (9/0041); G01L (1/103); G01L (27/002); G01L (3/1428); G01L (3/1478); G01L (5/13); G01L (9/0014); G01L (9/0079); G01L (9/10); G01L (1/167); G01L (11/04); G01L (19/0645); G01L (19/14); G01L (3/00); G01L (3/045); G01L (3/1435); G01L (9/0005); G01L (9/0013); G01L (9/0025); G01L (9/0048); G01L (9/0076); G01L (19/0007); G01L (19/083); G01L (3/14); G01L (3/1485); G01L (3/247); G01L (5/26); G01L (7/00); G01L (9/005); G01L (9/0064); G01L (11/00); G01L (11/06); G01L (17/005); G01L (19/003); G01L (27/005); G01L (5/03); G01L (9/02); G01L (1/046); G01L (11/006); G01L (17/00); G01L (19/0061); G01L (19/08); G01L (25/006); G01L (3/02); G01L (3/04); G01L (3/08); G01L (3/109); G01L (5/0023); G01L (5/169); G01L (7/024); G01L (7/16); G01L (9/0023); G01L (13/023); G01L (19/00); G01L (19/0046); G01L (23/08); G01L (23/22); G01L (3/1471); G01L (5/166); G01L (5/18); G01L (5/20); G01L (7/082); G01L (7/18); G01L (9/00); G01L (9/0004); G01L (9/0027); G01L (9/0047); G01L (9/14); G01L (9/18); G01L (1/048); G01L (1/08); G01L (1/083); G01L (1/086); G01L (11/008); G01L (19/0069); G01L (19/0618); G01L (19/0627); G01L (19/141); G01L (19/143); G01L (19/145); G01L (19/16); G01L (21/00); G01L (21/02); G01L (21/20); G01L (23/02); G01L (23/12); G01L (23/16); G01L (27/00); G01L (3/1414); G01L (3/1464); G01L (3/16); G01L (3/22); G01L (5/163); G01L (5/286); G01L (7/04); G01L (7/043); G01L (7/08); G01L (7/084); G01L (7/10); G01L (9/0007); G01L (9/0008); G01L (9/0017); G01L (9/0038); G01L (9/0044); G01L (9/0045); G01L (9/0082); G01L (9/0083); G01L (9/0086); G01L (9/0092); G01L (9/025); G01L (15/00); G01L (19/0023); G01L (19/0654); G01L (19/0681); G01L (19/069); G01L (19/142); G01L (23/222); G01L (3/1492); G01L (3/18); G01L (5/1623); G01L (5/171); G01L (5/173); G01L (7/02); G01L (7/022); G01L (7/166); G01L (9/003); G01L (9/0036); G01L (9/105); E01D (19/04)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4848478	12/1988	Hafner	73/1.13	G01L 25/00
2018/0238771	12/2017	Ehmke	N/A	B61K 9/08

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2415054	12/2000	CN	N/A
112748067	12/2020	CN	G01L 5/168
113155727	12/2020	CN	E01D 19/04
114166401	12/2021	CN	N/A
118310667	12/2023	CN	N/A
118424552	12/2023	CN	N/A
0270123	12/1987	EP	N/A
2012078286	12/2011	JP	G01L 25/00

Background/Summary

BACKGROUND OF THE PRESENT INVENTION

Field of Invention

(1) The present invention relates to the technical field of rail transit structural engineering, can further be applied to the structural engineering fields of highways, municipal administration, buildings, and the like, and particularly relates to a determination method for accurately determining a coefficient of friction between a force measurement apparatus and a bearing.

Description of Related Arts

(2) A force measurement apparatus and a bearing are used to determine a force applied to a force transmission member between structures, and are widely used in various fields, especially in the field of bridge engineering. The force measurement apparatus is used as a main force transmission member between an upper structure and a lower structure of a bridge structure, and a change in the force applied to the force measurement apparatus can reflect an overall operation situation of the bridge to a large extent, to implement acquisition of monitoring data of a bridge bearing, i.e., bridge vertical reaction, which can provide a technical basis for bridge health monitoring. As the construction of highway and railway bridges in China increases year by year, monitoring of vertical static loads and dynamic loads of bridge bearings is of great practical significance for operation of bridges.

(3) As for a wedge-shaped height-adjustment force measurement apparatus, since a wedge block is inevitably worn during operation, and considering aging performance of the material itself, a coefficient of friction of the wedge-shaped height-adjustment force measurement apparatus may change. When a vertical force is reversely deduced by measuring a horizontal force of a wedge-shaped mechanism of the force measurement apparatus, the change in the coefficient of friction affects the accuracy of the vertical force because the solution of the vertical force is related to the coefficient of friction. Therefore, the coefficient of friction needs to be determined on site.

However, currently, there is no mature solution to the problem of determining a coefficient of friction of a wedge-shaped force measurement apparatus.

SUMMARY OF THE PRESENT INVENTION

(4) An objective of the present invention is to provide a determination method for accurately determining a coefficient of friction between a force measurement apparatus and a bearing, to solve the problem of inaccurate measurement of a vertical force due to the fact that a coefficient of friction of a wedge-shaped height-adjustment force measurement apparatus in service cannot be accurately determined.

(5) To solve the above technical problem, a solution adopted by the present invention is as follows: a method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction, the method comprising the following steps: A1: recording a value of a horizontal or lateral force applied to a force measurement apparatus, this step comprising two situations of a1 and a2: a1: first recording a horizontal or lateral force value of a sensing apparatus before the force measurement apparatus is lifted; and then applying, by a power apparatus, a horizontal or lateral force to two wedge-shaped load transducers on the force measurement apparatus to change a height of the force measurement apparatus, and recording a horizontal or lateral force value of the sensing apparatus during the change in height or during a pause in the change; a2: first applying a horizontal or lateral force to two wedge-shaped load transducers on the force measurement apparatus to change a height of the force measurement

apparatus, and recording a horizontal or lateral force value of a sensing apparatus during the change in height or during a pause in the change; then accurately adjusting a gap between the two load transducers, to tightly attach the sensing apparatus to the load transducers; and releasing a power apparatus, and recording a horizontal or lateral force value measured by the sensing apparatus between the two load transducers at this time; and a3: first applying a horizontal or lateral force to two wedge-shaped load transducers to change a height of the force measurement apparatus, and recording a horizontal or lateral force value of a sensing apparatus during the change in height or during a pause in the change; and then reducing the value of the horizontal or lateral force applied to the two wedge-shaped load transducers to change the height of the force measurement apparatus, and recording a horizontal or lateral force value of the sensing apparatus during the change in height or during a pause in the change; A2: deriving a relational expression of a coefficient of friction, an angle of inclination of the load transducers and a horizontal or lateral force according to a mechanical equilibrium relationship between forces applied to the load transducers in a1 or a2 or a3; and A3: calculating a value of the coefficient of friction according to the horizontal or lateral force values recorded in a1 or a2 or a3 and the relational expression in A2.

(6) Preferably, the method further comprises step A4: deriving a relational expression of a vertical force, the angle of inclination of the load transducers, the horizontal or lateral force and the coefficient of friction according to the mechanical equilibrium relationship between the forces applied to the load transducers in a1 or a2 or a3, and accurately calculating a vertical force value of the force measurement apparatus in combination with the value of the coefficient of friction obtained in A3.

(7) Preferably, in a1, a2 and a3, the force measurement apparatus used comprises an upper adjusting plate and a lower adjusting plate, a bearing core is arranged above the upper adjusting plate or below the lower adjusting plate, a bottom surface of the upper adjusting plate is provided with an adjusting chamber, the two wedge-shaped load transducers are arranged in the adjusting chamber, one or more sensing apparatuses are arranged between the two load transducers, the force measurement apparatus is provided with the power apparatus, an output end of the power apparatus is connected in series to the sensing apparatus(es) and is then in contact with a side portion and/or an end portion of one of the load transducers, a fixed end of the power apparatus is connected to a side portion and/or an end portion of the other load transducer, the power apparatus outputs a horizontal or lateral force to push the two load transducers away from each other, so that power output is paused when a height of the upper adjusting plate rises to a set elevation, and at this time, the value of the horizontal or lateral force applied by the power apparatus is capable of being directly measured by the sensing apparatus(es) connected in series to the output end.

(8) Preferably, a limiting apparatus(es) is/are provided on one side or two sides of each sensing apparatus, each limiting apparatus is an adjusting screw, and the adjusting screw is in threaded connection with the corresponding load transducer; in step a2, after the power apparatus relatively moves the two load transducers away each other, the height of the force measurement apparatus rises to a set elevation, and a gap is formed between the two load transducers; and by means of the adjusting screw, the sensing apparatus is adjusted to be tightly attached to the load transducers, and after the horizontal or lateral force applied by the power apparatus is released, the sensing apparatus between the two load transducers accurately measures the horizontal or lateral force value at this time.

(9) Preferably, a locking apparatus is arranged between the sensing apparatus and the adjusting screw, and the locking apparatus is a steel plate, section steel, round steel, angle steel, a steel pipe, or a concrete block. An appropriate number of locking apparatuses may be selected based on an actual situation on site, so that the locking apparatuses are adjustable and easy to operate.

(10) Preferably, two ends of the sensing apparatus are provided with matching fixing seats, the fixing seats are located between the sensing apparatus and the limiting apparatus, and one or more groups of fixing seats are provided.

(11) Preferably, the limiting apparatus is two wedge blocks, inclined surfaces of the two wedge blocks match and are attached to each other, and by means of mutual sliding of the two wedge blocks, the sensing apparatus is adjusted to be tightly attached to the load transducers. The two wedge blocks may accurately adjust a gap between the sensing apparatus and the load transducer by sliding up and down or back and forth relative to each other, so that the sensing apparatus is tightly attached to the load transducer. By using a lever principle of the wedge blocks, a larger transverse force (which force is in a horizontal direction) can be obtained from a smaller force (which force is in the same direction as the sliding direction of the two wedge blocks), thereby achieving the purpose of tightly attaching the sensing apparatus to the load transducer, making the adjustment faster and saving time and labor.

(12) Preferably, a1 in A1 is implemented by means of another structure in which the force measurement apparatus comprises a bearing core, a lower seat plate, and two load transducers, wherein a bottom surface of the bearing core is in contact with top surfaces of the two load transducers in the form of an oblique straight surface or a cylindrical surface or a curved surface, a plurality of sensing apparatuses are arranged between each load transducer and a side wall of the lower seat plate in a horizontal direction, and the sensing apparatuses record horizontal or lateral force values.

(13) Preferably, a power apparatus is installed on either side wall of the lower seat plate, and an output end of the power apparatus passes through the side wall of the lower seat plate and is in direct or indirect contact with the sensing apparatuses; the power apparatus applies a horizontal or lateral force such that the sensing apparatuses are pressed and the two load transducers move close to each other to jack up the bearing core; a value of the horizontal or lateral force applied to the sensing apparatuses is recorded when a height of the bearing core starts to change or the change is paused; a load distribution baffle is arranged between the output end of the power apparatus and the sensing apparatuses, the output end of the power apparatus is in contact with a side face of the load distribution baffle, and the sensing apparatuses are located between the load distribution baffle and the respective load transducer; and an adjustable limiting block is installed between the load distribution baffle and the side wall of the lower seat plate.

(14) Preferably, the bearing core is placed in a bottom basin having a basin cavity structure, a force measurement carrier is arranged between the bearing core and the bottom basin, a sensing apparatus is arranged on a side face of the force measurement carrier, and the sensing apparatus is connected to an external data acquisition system via a signal line.

(15) Preferably, the top surfaces of the load transducers are in contact with a top surface of the adjusting chamber in the form of oblique straight surfaces or cylindrical surfaces or curved surfaces, the bottom surfaces of the load transducers are in contact with an upper supporting surface of the lower adjusting plate in the form of planar or curved surfaces or oblique straight surfaces, a matching upper friction pair and lower friction pair are respectively arranged between the top surface of the adjusting chamber and the top surfaces of the load transducers, and between the bottom surfaces of the load transducers and the top surface of the lower adjusting plate, and each power apparatus is a hydraulic cylinder, a pneumatic cylinder or a mechanical transmission mechanism.

(16) The present invention has the following beneficial effects. 1. According to the present invention, a relational expression of a coefficient of friction, a horizontal or lateral force and an angle of inclination of the load transducers is derived according to a mechanical equilibrium relationship between forces applied to the load transducers, and the coefficient of friction is accurately obtained. A vertical force is accurately calculated according to a derived relational expression of the vertical force, the angle of inclination of the load transducers, the horizontal or lateral force and the coefficient of friction. According to this method, the coefficient of friction can be accurately determined when the force measurement apparatus and the bearing are in service, so that the vertical force can be accurately calculated, and the vertical force of the bearing is quickly

and accurately calibrated, thereby ensuring accurate monitoring of the force applied to the bearing during an operation period, and improving the safety of the bridge structure. 2. According to this method, the gap between the two load transducers can be accurately adjusted by means of an adjusting screw or wedge blocks, so that a sensing apparatus is tightly attached to the load transducers, and thus the height of the apparatus does not change in the two equilibrium relational expressions. The unchanged height means that a load on an upper portion does not change, so that the value of the coefficient of friction calculated based on the unchanged height of the apparatus is more accurate, and the calibration of the vertical force is also more accurate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic flow structure diagram of the present invention.
- (2) FIG. 2 is a schematic structural diagram of the present invention.
- (3) FIG. 3 is a schematic structural diagram, with a bearing core being located above an upper adjusting plate.
- (4) FIG. 4 is a schematic structural diagram of series connection of an adjusting screw and a sensor.
- (5) FIG. 5 is a schematic structural diagram, with a locking apparatus being provided.
- (6) FIG. 6 is another schematic structural diagram of a locking apparatus and an adjusting screw.
- (7) FIG. 7 is a schematic structural diagram, with fixing seats being provided.
- (8) FIG. 8 is a structural diagram, with a limiting apparatus being wedge blocks.
- (9) FIG. 9 is a structural diagram, with a power apparatus being added to FIG. 8.
- (10) FIG. 10 is a schematic diagram of another sliding structure of two wedge blocks.
- (11) FIG. 11 is a schematic structural diagram, with wedge blocks being connected to a power apparatus.
- (12) FIG. 12 is a structural diagram, with a power apparatus being connected in FIG. 2.
- (13) FIG. 13 is a schematic structural diagram in Embodiment 9.
- (14) FIG. 14 is a structural diagram, with three power apparatuses being provided in Embodiment 9.
- (15) FIG. 15 is a structural diagram, with one power apparatus being provided in Embodiment 9.
- (16) FIG. 16 is a structural diagram, with two power apparatuses being provided in Embodiment 9.
- (17) FIG. 17 is a free-body diagram of forces applied to a load transducer, with a height of a force measurement apparatus dropping.
- (18) FIG. 18 is a free-body diagram of forces applied to a load transducer, with a force measurement apparatus rising at a high speed.
- (19) FIG. 19 is a schematic structural diagram of Embodiment 10.

LIST OF REFERENCE NUMERALS

- (20) 1—bearing core, 2—upper friction pair, 3—load transducer, 4—lower friction pair, 5—sensing apparatus, 6—upper adjusting plate, 7—bolt, 8—lower adjusting plate, 9—adjusting chamber, 10—adjusting screw, 11—locking apparatus, 12—fixing seat, 13—power apparatus, 14—wedge block, 15—lower seat plate, 16—adjustable limiting block, 17—load distribution baffle, 18—bottom basin, 19—force measurement carrier, 20—data acquisition system.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(21) The present invention will be further described in detail below with reference to embodiments and the accompanying drawings, but implementations of the present invention are not limited thereto.

(22) In the description of the present invention, it should be noted that an orientation or positional relationship indicated by the term “centre”, “upper”, “lower”, “left”, “right”, “vertical”, “longitudinal”, “lateral”, “horizontal”, “inner”, “outer”, “front”, “rear”, “top”, “bottom” or the like

is based on the orientation or positional relationship shown in the accompanying drawings, or the orientation or positional relationship in which a product of the present invention is conventionally placed when in use, and are intended to facilitate the description of the present invention and simplify the description only, rather than indicating or implying that the device or element referred to must have a particular orientation or be constructed and operated in a particular orientation, and will not to be interpreted as limiting the present invention.

(23) In the description of the present invention, it should be further noted that unless otherwise explicitly specified and defined, the terms “arrange”, “provided with”, “install”, “connected”, and “connection” should be understood in a broad sense and, for example, may be a fixed connection, a detachable connection, or an integrated connection; or may be a mechanical connection or an electrical connection; or may be a direct connection, an indirect connection by means of an intermediate medium, or internal communication between two elements. For a person of ordinary skill in the art, the specific meanings of the terms mentioned above in the present invention should be construed according to specific circumstances.

Embodiment 1

(24) A method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction comprises the following steps.

(25) In step A1, a value of a horizontal or lateral force applied to a force measurement apparatus is recorded, this step comprising two methods of a1 and a2.

(26) In the first method a1, a horizontal or lateral force value $T_{\text{sub.Drop}}$ of a sensing apparatus 5 before the force measurement apparatus is lifted is first recorded. A power apparatus 13 then applies a horizontal or lateral force to two wedge-shaped load transducers 3 on the force measurement apparatus to change a height of the force measurement apparatus, the two load transducers 3 move away from each other, the height of the force measurement apparatus rises, and a horizontal or lateral force value $T_{\text{sub.Rise}}$ of the sensing apparatus 5 during the change in height or during a pause in the change is recorded.

(27) In the second method a2, a power apparatus 13 first applies a horizontal or lateral force to two wedge-shaped load transducers 3 on the force measurement apparatus, the two load transducers 3 move away from each other to rise the height of the force measurement apparatus, and a horizontal or lateral force value $T_{\text{sub.Rise}}$ of the sensing apparatus 5 during the change in height or during a pause in the change is recorded. A gap between the two load transducers 3 is then accurately adjusted, to tightly attach the sensing apparatus 5 to the load transducers 3. The power apparatus 13 is released, and a horizontal or lateral force value $T_{\text{sub.Drop}}$ measured by the sensing apparatus 5 between the two load transducers 3 is recorded at this time.

(28) In the third method a3, a horizontal or lateral force is first applied to two wedge-shaped load transducers 3 to rise a height of the force measurement apparatus, and a horizontal or lateral force value $T_{\text{sub.Rise}}$ of a sensing apparatus 5 during the change in height or during a pause in the change is recorded. The value of the horizontal or lateral force applied to the two wedge-shaped load transducers 3 is then reduced to drop the height of the force measurement apparatus, and a horizontal or lateral force value $T_{\text{sub.Drop}}$ of the sensing apparatus 5 during the change in height or during a pause in the change is recorded.

(29) A2: A relational expression of a coefficient of friction, an angle of inclination of the load transducers 3 and a horizontal or lateral force is derived according to a mechanical equilibrium relationship between forces applied to the load transducers in a1 or a2 or a3.

(30) In step A3, a value of the coefficient of friction is calculated according to the horizontal or lateral force values recorded in a1 or a2 or a3 and the relational expression in A2.

(31) Specifically, in a1, a vertical force applied to the force measurement apparatus is F , vertical forces transmitted to the two wedge-shaped transducers 3 from this vertical force are respectively $F/2$, friction forces applied to upper and lower sliding surfaces of the load transducers 3 are respectively f_1 and f_2 , the upper sliding surface is subjected to a pressing force N_1 perpendicular

thereto, the lower sliding surface is subjected to a vertical pressing force $F/2$, the forces applied to the load transducers **3** are decomposed, and the following is obtained according to a mechanical equilibrium relationship: $f_1 = N_{\text{sub.1}} \times \mu_{\text{sub.1}}$, $f_2 = \mu_{\text{sub.2}} \times F/2$, and $N_{\text{sub.1}} = F/2(\cos \theta + \mu_{\text{sub.1}} \times \sin \theta)$, wherein an equivalent coefficient of friction μ replaces the coefficients of friction $\mu_{\text{sub.1}}$ and $\mu_{\text{sub.2}}$ of the upper and lower sliding surfaces to obtain

$$(32) T_{\text{Drop}} = 0.5F \cdot \text{Math.} \frac{\tan^2 - 2 - \tan^2}{1 + \tan^2}.$$

After the power apparatus **13** applies a horizontal force, friction forces applied to the upper and lower sliding surfaces of the load transducers are respectively f_3 and f_4 , the upper sliding surface is subjected to a pressing force N_2 perpendicular thereto, the lower sliding surface is subjected to a vertical pressing force $F/2$, the forces applied to the load transducers **3** are decomposed, and the following is obtained according to a mechanical equilibrium relationship: $f_3 = N_{\text{sub.1}} \times \mu_{\text{sub.1}}$, $f_4 = \mu_{\text{sub.2}} \times F/2$, and $N_{\text{sub.2}} = F/2(\cos \theta - \mu_{\text{sub.1}} \times \sin \theta)$, wherein an equivalent coefficient of friction μ replaces the coefficients of friction $\mu_{\text{sub.1}}$ and $\mu_{\text{sub.2}}$ of the upper and lower sliding surfaces to obtain

$$(33) T_{\text{Rise}} = \frac{F}{2} \cdot \text{Math.} \frac{2 + \tan^2 - \tan^2}{1 - \tan^2} |$$

(θ is an angle of inclination of the load transducers).

$$(34) = \frac{T_{\text{Rise}} - T_{\text{Drop}}}{T_{\text{Rise}} + T_{\text{Drop}}} \cdot \text{Math.} \frac{\tan^2}{2 + \tan^2}$$

can be obtained based on the two formulas of $T_{\text{sub.Rise}}$ and $T_{\text{sub.Drop}}$, $T_{\text{sub.Rise}}$ and $T_{\text{sub.Drop}}$ are directly obtained by the sensing apparatus **5**, and the value of the coefficient of friction can be accurately calculated.

(35) In a_2 , the power apparatus **13** first applies a horizontal force to move the two load transducers **3** away from each other, the height of the force measurement apparatus rises, and the horizontal force value $T_{\text{sub.Rise}}$ is recorded at this time; friction forces applied to upper and lower sliding surfaces of the load transducers **3** are respectively f_3 and f_4 , the upper sliding surface is subjected to a pressing force N_2 perpendicular thereto, the lower sliding surface is subjected to a vertical pressing force $F/2$, the forces applied to the load transducers **3** are decomposed, and the following is obtained according to a mechanical equilibrium relationship: $f_3 = N_{\text{sub.1}} \times \mu_{\text{sub.1}}$, $f_4 = \mu_{\text{sub.2}} \times F/2$, and $N_{\text{sub.2}} = F/2(\cos \theta - \mu_{\text{sub.1}} \times \sin \theta)$, wherein an equivalent coefficient of friction μ replaces the coefficients of friction $\mu_{\text{sub.1}}$ and $\mu_{\text{sub.2}}$ of the upper and lower sliding surfaces to obtain

$$(36) T_{\text{Rise}} = \frac{F}{2} \cdot \text{Math.} \frac{2 + \tan^2 - \tan^2}{1 - \tan^2} |$$

(θ is an angle of inclination of the load transducers). Since the height of the force measurement apparatus rises to a set elevation after the two load transducers **3** relatively move away from each other, a gap is formed between the two load transducers **3**. After the power apparatus **13** is released, a vertical load of an upper member completely acts on the force measurement apparatus. A limiting apparatus is used for adjustment to eliminate the gap between two load transducers **3**. After the power apparatus **13** is released, relative displacement will not occur again because there is no gap between the two load transducers **3**. At this time, the height of the force measurement apparatus is kept unchanged, and the sensing apparatus **5** between the two load transducers **3** measures the horizontal or lateral force value $T_{\text{sub.Drop}}$ more accurately. At this time, friction forces applied to the upper and lower sliding surfaces of the load transducers **3** are respectively f_1 and f_2 , the upper sliding surface is subjected to a pressing force N_1 perpendicular thereto, the lower sliding surface is subjected to a vertical pressing force $F/2$, the forces applied to the load transducers **3** are decomposed, and the following is obtained according to a mechanical equilibrium relationship: $f_1 = N_{\text{sub.1}} \times \mu_{\text{sub.1}}$, $f_2 = \mu_{\text{sub.2}} \times F/2$, and $N_{\text{sub.1}} = F/2(\cos \theta + \mu_{\text{sub.1}} \times \sin \theta)$, wherein an equivalent coefficient of friction μ replaces the coefficients of friction $\mu_{\text{sub.1}}$ and $\mu_{\text{sub.2}}$ of the upper and lower sliding surfaces to obtain

$$(37) T_{\text{Drop}} = 0.5F \cdot \text{Math.} \frac{\tan^2 - 2 - \tan^2}{1 + \tan^2}, = \frac{T_{\text{Rise}} - T_{\text{Drop}}}{T_{\text{Rise}} + T_{\text{Drop}}} \cdot \text{Math.} \frac{\tan^2}{2 + \tan^2}$$

can be obtained based on the two formulas of T.sub.Rise and T.sub.Drop, T.sub.Rise and T.sub.Drop are directly obtained by the sensing apparatus 5, and the value of the coefficient of friction can be accurately calculated.

(38) The mechanical equilibrium relational expression in a3 is the same as those in a1 and a2.

Embodiment 2

(39) The method further comprises step A4, in which a relational expression of a vertical force, the angle of inclination of the load transducers 3, the horizontal or lateral force and the coefficient of friction is derived according to the mechanical equilibrium relationship between the forces applied to the load transducers 3 in a1 or a2 or a3, and a vertical force value of the force measurement apparatus is accurately calculated in combination with the value of the coefficient of friction obtained in A3.

(40) Specifically,

$$(41) F = 2T_{\text{Drop}} \cdot \text{Math.} \frac{1 + \tan}{\tan - 2 - 2 \tan}$$

is obtained through the conversion of the relational expression of T.sub.Drop, the calculated value of the coefficient of friction and the known horizontal force value are substituted into this relational expression, so that the accurate vertical force value is obtained, and the vertical force can be calibrated by means of the horizontal force.

Embodiment 3

(42) In a1, a2, and a3, the force measurement apparatus used comprises an upper adjusting plate 6 and a lower adjusting plate 8, a bearing core 1 is arranged above the upper adjusting plate 6 or below the lower adjusting plate 8, a bottom surface of the upper adjusting plate 6 is provided with an adjusting chamber 9, and the two wedge-shaped load transducers 3 are arranged in the adjusting chamber 9. The top surfaces of the load transducers 3 are in contact with a top surface of the adjusting chamber 9 in the form of oblique straight surfaces or cylindrical surfaces or curved surfaces, the bottom surfaces of the load transducers 3 are in contact with an upper supporting surface of the lower adjusting plate 8 in the form of planar or curved surfaces or oblique straight surfaces, and a matching upper friction pair 2 and lower friction pair 4 are respectively arranged between the top surface of the adjusting chamber 9 and the top surfaces of the load transducers 3, and between the bottom surfaces of the load transducers 3 and the top surface of the lower adjusting plate. One or more sensing apparatuses 5 are arranged between the two load transducers 3, the force measurement apparatus is provided with a power apparatus 13, and the power apparatus 13 is a hydraulic cylinder, a pneumatic cylinder or a mechanical transmission mechanism. A hydraulic jack is used in this embodiment. An output end of the power apparatus 13 is connected in series to the sensing apparatus(es) 5 and is then in contact with a side portion and/or an end portion of one of the load transducers 3, a fixed end of the power apparatus 13 is connected to a side portion and/or an end portion of the other load transducer 3, the power apparatus 13 outputs a horizontal or lateral force to push the two load transducers 3 away from each other, so that power output is paused when a height of the upper adjusting plate 6 rises to a set elevation, and at this time, the value of the horizontal or lateral force applied by the power apparatus 13 can be directly measured by the sensing apparatus(es) 5 connected in series to the output end.

Embodiment 4

(43) A limiting apparatus(es) is/are provided on one side or two sides of each sensing apparatus 5, and each limiting apparatus can eliminate the gap between the two load transducers 3, and ensure that the height of the force measurement apparatus will not change again after the force measurement apparatus reaches a set elevation. The limiting apparatus is an adjusting screw 10, and the adjusting screw 10 is in threaded connection with the corresponding load transducer 3. In step a2, after the power apparatus 13 relatively moves the two load transducers 3 away each other, the height of the force measurement apparatus rises to a set elevation, and a gap is formed between the two load transducers 3. By means of the adjusting screw 10, the sensing apparatus 5 is adjusted

to be tightly attached to the load transducers **3**, and after the horizontal or lateral force applied by the power apparatus **13** is released, the sensing apparatus **5** between the two load transducers **3** accurately measures the horizontal or lateral force value at this time.

Embodiment 5

(44) A locking apparatus **11** is arranged between the sensing apparatus **5** and the adjusting screw **10**, and the locking apparatus **11** is a steel plate, section steel, round steel, angle steel, a steel pipe, or a concrete block. An appropriate number of locking apparatuses **11** may be selected based on an actual situation on site, so that the locking apparatuses **11** are adjustable and easy to operate.

(45) This embodiment further has another structure, in which two opposite ends of the two load transducers **3** are in threaded connection with adjusting screws **10**, locking apparatuses **11** are respectively connected in series between two opposite end portions of the two adjusting screws **10**, and a sensing apparatus **5** is arranged between two locking apparatuses **11**.

Embodiment 6

(46) In this embodiment, one end or two ends of each adjusting screw **10** is/are in threaded connection with the corresponding load transducer(s) **3**, and a sensing apparatus is arranged between two adjusting screws **10** or connected in series to the free end of the adjusting screw **10**.

Embodiment 7

(47) Two ends of the sensing apparatus **5** are provided with matching fixing seats **12**, the fixing seats **12** are located between the sensing apparatus **5** and the limiting apparatus, and one or more groups of fixing seats **12** are provided. That is, the fixing seats **12** may be separated or integrated. If the fixing seats are separated, each sensor apparatus **5** corresponds to one group of fixing seats **12**. If the fixing seats are integrated, a plurality of sensing apparatuses **5** share one group of fixing seats **12**.

Embodiment 8

(48) In this embodiment, the limiting apparatus is in the form of two wedge blocks **14**, inclined surfaces of the two wedge blocks **14** match and are attached to each other, and by means of mutual sliding of the two wedge blocks **14**, the sensing apparatus **5** is adjusted to be tightly attached to the load transducers **3**. The two wedge blocks **14** may accurately adjust a gap between the sensing apparatus **5** and the load transducer **3** by sliding up and down or back and forth relative to each other, so that the sensing apparatus **5** is tightly attached to the load transducer **6**. By using a lever principle of the wedge blocks **14**, a larger transverse force (which force is in a horizontal direction) can be obtained from a smaller force (which force is in the same direction as the sliding direction of the two wedge blocks **14**), thereby achieving the purpose of tightly attaching the sensing apparatus **5** to the load transducer **13**, making the adjustment faster and saving time and labor. On this basis, the two wedge blocks **14** may be further provided with power apparatuses **13**, an output end of each power apparatus **13** is in contact with an end portion of one of the wedge blocks **14**, and a fixed end of the power apparatus **13** is connected to the other wedge block **14**. The power apparatus **13** applies a force to play an auxiliary role. In addition, the power apparatus **13** may be further connected to an external controller to implement intelligent adjustment, which is more efficient and faster.

(49) In a2, after the horizontal or lateral force applied by the power apparatus **13** is released, a vertical load of an upper member (beam body) completely acts on the force measurement apparatus. Since the sensing apparatus **5** is adjusted to be tightly attached to the load transducers **3** by means of the wedge blocks **14**, there is no gap between the two load transducers **3**. Under the vertical load on the force measurement apparatus, relative displacement between the two load transducers **3** will not occur again. At this time, the height of a height-adjustment apparatus is kept unchanged, and the sensing apparatus **5** between the two load transducers **3** measures the horizontal or lateral force value more accurately.

Embodiment 9

(50) For a1 in A1, another structure is adopted in this embodiment to implement the method of the

present invention. In this embodiment, a contact surface between the bearing core **1** and the load transducer **3** is an oblique straight surface, and a bottom surface of the bearing core **1** is configured to have a V shape. A contact surface between the load transducer **3** and a lower seat plate **15** is a planar surface. A plurality of sensing apparatuses **5** are located between the load transducer **3** and a side wall of the lower seat plate **15** in a horizontal direction, and the lower seat plate **15** is of a basin cavity structure. A vertical force applied to the bearing core **1** is decomposed by the load transducer **3** into horizontal components, so that the sensing apparatuses **5** are pressed, and the sensing apparatuses **5** record the horizontal or lateral force values at this time.

(51) A power apparatus **13** is installed on either side wall of the lower seat plate **15**, and an output end of the power apparatus **13** passes through the side wall of the lower seat plate **15** and is in direct or indirect contact with the sensing apparatuses **5**. The power apparatus **13** applies a horizontal force such that the sensing apparatuses **5** are pressed. The power apparatus **13** may implement power output in the form of a hydraulic cylinder or a pneumatic cylinder. The two load transducers **3** move close to each other to jack up the bearing core **1**. A value of the horizontal or lateral force applied to the sensing apparatuses **5** is recorded when a height of the bearing core **1** starts to change or the change is paused. A relational expression of a coefficient of friction, an angle of inclination of the load transducers **3** and a horizontal or lateral force is derived according to a mechanical equilibrium relationship between forces applied to the wedge-shaped load transducers **3**, and a value of the coefficient of friction is calculated according to the relational expression.

(52) A load distribution baffle **17** is arranged between the output end of the power apparatus **13** and the sensing apparatuses **5**, the output end of the power apparatus **13** is in contact with a side face of the load distribution baffle **17**, and the sensing apparatuses **5** are located between the load distribution baffle **17** and the respective load transducer **3**. With the load distribution baffle **17**, the number of power apparatuses **13** used can be reduced. In this embodiment, three sensing apparatuses **5** are provided, and three power apparatuses **13** are correspondingly used. It is also possible that one or two power apparatuses **13** correspond to the three sensing apparatuses **5**. An adjustable limiting block **16** is installed between the load distribution baffle **17** and the side wall of the lower seat plate **15**.

Embodiment 10

(53) As shown in FIG. **19**, in this embodiment, the bearing core is placed in a bottom basin having a basin cavity **18** structure, a force measurement carrier **19** is arranged between the bearing core **1** and the bottom basin **18**, and a sensing apparatus **5** is arranged on a side face of the force measurement carrier **19**. A signal line of the sensing apparatus **5** passes through a mounting hole in a side wall of the bottom basin **18** and is connected to an external data acquisition system **20**, and the data acquisition system **20** is configured to monitor a vertical supporting force of a support apparatus. An electric signal generated when the sensing apparatus **5** is subjected to a horizontal pressing force is transmitted through a wireless or wired network to the data acquisition system **20** for collection, analysis and processing to measure the load, and a vertical load is obtained reversely through mechanical equilibrium decomposition on the load transducers **3**, so as to achieve the purpose of force measurement. The data acquisition system **20** includes a demodulator and a computer system (known in the prior art). An input end of the demodulator is connected, via a signal line, to the sensing apparatus **5** installed on the force measurement carrier **19**, and an output end of the demodulator is connected to the computer system through a wireless network or a wired network. The computer system is further provided with a real-time monitoring and alarm system. The sensing apparatus **5** senses a pressure stress change of the force measurement carrier **19** and outputs a pressure wavelength signal to the demodulator by means of the signal line. After analyzing and processing the wavelength signal, the demodulator transmits the signal through the wireless network or the wired network to the computer system for processing, and then an actual load of the bearing is visually displayed and compared with a normal service load of the bearing. An alarm is given in the case of an abnormality, so as to monitor the load of the bearing from a

long distance, and data of the force applied to the bearing is acquired in real time.

(54) Those described above are only the preferred embodiments of the present invention, and are not intended to limit the present invention in any form. Any simple modifications, equivalent replacements, improvements, etc. made to the above embodiments within the spirit and principle of the present invention based on the technical essence of the present invention still fall within the scope of protection of the technical solutions of the present invention.

Claims

1. A method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction, characterized by comprising the following steps: A1: recording a value of a horizontal or lateral force applied to a force measurement apparatus, this step comprising three situations of a1, a2, and a3: a1: first recording a horizontal or lateral force value of a sensing apparatus (5) before the force measurement apparatus is lifted; and then applying, by a power apparatus (13), a horizontal or lateral force to two wedge-shaped load transducers (3) on the force measurement apparatus to change a height of the force measurement apparatus, and recording a horizontal or lateral force value of the sensing apparatus (5) during the change in height or during a pause in the change; a2: first applying a horizontal or lateral force to two wedge-shaped load transducers (3) on the force measurement apparatus to change a height of the force measurement apparatus, and recording a horizontal or lateral force value of a sensing apparatus (5) during the change in height or during a pause in the change; then accurately adjusting a gap between the two load transducers (3), to tightly attach the sensing apparatus (5) to the load transducers (3); and releasing a power apparatus (13), and recording a horizontal or lateral force value measured by the sensing apparatus (5) between the two load transducers (3) at this time; and a3: first applying a horizontal or lateral force to two wedge-shaped load transducers (3) to change a height of the force measurement apparatus, and recording a horizontal or lateral force value of a sensing apparatus (5) during the change in height or during a pause in the change; and then reducing the value of the horizontal or lateral force applied to the two wedge-shaped load transducers (3) to change the height of the force measurement apparatus, and recording a horizontal or lateral force value of the sensing apparatus (5) during the change in height or during a pause in the change; A2: deriving a relational expression of a coefficient of friction, an angle of inclination of the load transducers (3) and a horizontal or lateral force according to a mechanical equilibrium relationship between forces applied to the load transducers in a1 or a2 or a3; and A3: calculating a value of the coefficient of friction according to the horizontal or lateral force values recorded in a1 or a2 or a3 and the relational expression in A2.

2. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 1, characterized by further comprising step A4: deriving a relational expression of a vertical force, the angle of inclination of the load transducers (3), the horizontal or lateral force and the coefficient of friction according to the mechanical equilibrium relationship between the forces applied to the load transducers (3) in a1 or a2 or a3, and accurately calculating a vertical force value of the force measurement apparatus in combination with the value of the coefficient of friction obtained in A3.

3. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 1, characterized in that in a1 and a2, the force measurement apparatus used comprises an upper adjusting plate (6) and a lower adjusting plate (8), a bearing core (1) is arranged above the upper adjusting plate (6) or below the lower adjusting plate (8), a bottom surface of the upper adjusting plate (6) is provided with an adjusting chamber (9), the two wedge-shaped load transducers (3) are arranged in the adjusting chamber (9), one or more sensing apparatuses (5) are arranged between the two load transducers (3), the force measurement apparatus is provided with the power apparatus (13), an output end of

the power apparatus (13) is connected in series to the sensing apparatus(es) (5) and is then in contact with a side portion and/or an end portion of one of the load transducers (3), a fixed end of the power apparatus (13) is connected to a side portion and/or an end portion of the other load transducer (3), the power apparatus (13) outputs a horizontal or lateral force to push the two load transducers (3) away from each other, so that power output is paused when a height of the upper adjusting plate (6) rises to a set elevation, and at this time, the value of the horizontal or lateral force applied by the power apparatus (13) is capable of being directly measured by the sensing apparatus(es) (5) connected in series to the output end.

4. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 3, characterized in that a limiting apparatus(es) is/are provided on one side or two sides of each sensing apparatus, each limiting apparatus is an adjusting screw (10), and the adjusting screw (10) is in threaded connection with the corresponding load transducer (3); in step a2, after the power apparatus (13) relatively moves the two load transducers (3) away each other, the height of the force measurement apparatus rises to a set elevation, and a gap is formed between the two load transducers (3); and by means of the adjusting screw (10), the sensing apparatus (5) is adjusted to be tightly attached to the load transducers (3), and after the horizontal or lateral force applied by the power apparatus (13) is released, the sensing apparatus (5) between the two load transducers (3) accurately measures the horizontal or lateral force value at this time.

5. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 3, characterized in that a locking apparatus (11) is arranged between the sensing apparatus (5) and the adjusting screw (10), and the locking apparatus (11) is a steel plate, section steel, round steel, angle steel, a steel pipe, or a concrete block.

6. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 3, characterized in that two ends of the sensing apparatus (5) are provided with matching fixing seats (12), the fixing seats (12) are located between the sensing apparatus (5) and the limiting apparatus, and one or more groups of fixing seats (12) are provided.

7. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 3, characterized in that the limiting apparatus is two wedge blocks (14), inclined surfaces of the two wedge blocks (14) match and are attached to each other, and by means of mutual sliding of the two wedge blocks (14), the sensing apparatus (5) is adjusted to be tightly attached to the load transducers (3).

8. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 3, characterized in that the bearing core (1) is placed in a bottom basin (18) having a basin cavity structure, a force measurement carrier (19) is arranged between the bearing core (1) and the bottom basin (18), a sensing apparatus (5) is arranged on a side face of the force measurement carrier (19), and the sensing apparatus (5) is connected to an external data acquisition system (20) via a signal line.

9. The method for calibrating an in-service bridge force-measurement bearing on the basis of accurate determination of a coefficient of friction according to claim 1, characterized in that a1 in A1 is implemented by means of another structure in which the force measurement apparatus comprises a bearing core (1), a lower seat plate (15), and two load transducers (3), wherein a bottom surface of the bearing core (1) is in contact with top surfaces of the two load transducers (3) in the form of an oblique straight surface or a cylindrical surface or a curved surface, a plurality of sensing apparatuses (5) are arranged between each load transducer (3) and a side wall of the lower seat plate (15) in a horizontal direction, and the sensing apparatuses (5) record horizontal or lateral force values.

10. The method for calibrating an in-service bridge force-measurement bearing on the basis of

accurate determination of a coefficient of friction according to claim 8, characterized in that a power apparatus (13) is installed on either side wall of the lower seat plate (15), and an output end of the power apparatus (13) passes through the side wall of the lower seat plate (15) and is in direct or indirect contact with the sensing apparatuses (5); the power apparatus (13) applies a horizontal or lateral force such that the sensing apparatuses (5) are pressed and the two load transducers (3) move close to each other to jack up the bearing core (1); a value of the horizontal or lateral force applied to the sensing apparatuses (5) is recorded when a height of the bearing core (1) starts to change or the change is paused; a load distribution baffle (17) is arranged between the output end of the power apparatus (13) and the sensing apparatuses (5), the output end of the power apparatus (13) is in contact with a side face of the load distribution baffle (17), and the sensing apparatuses are located between the load distribution baffle (17) and the respective load transducer (3); and an adjustable limiting block (16) is installed between the load distribution baffle (17) and the side wall of the lower seat plate (15).
